

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Into Operating Systems



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features.

Understands the concepts of server, its characteristics and its relation with operating systems.

Can Do Statements:

- I can talk about multi tasking operating systems.
- I can describe multi processor operating systems.
- I can explain what distributed operating systems are.

Ask your partner:

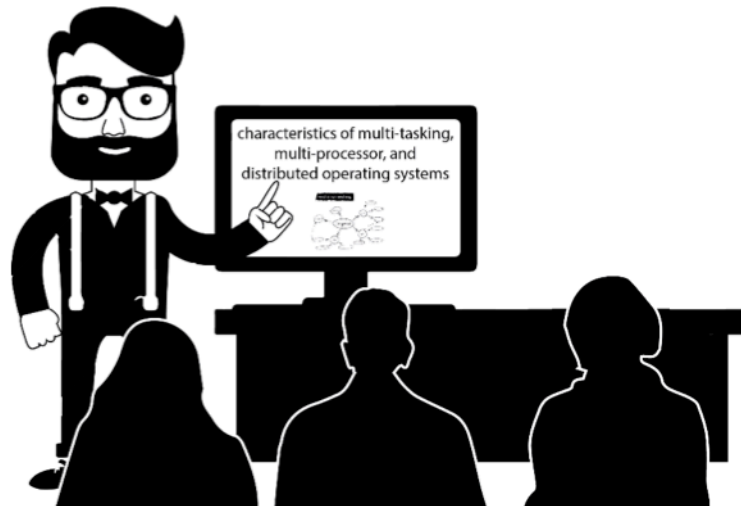
- What types of Operating Systems do you know?
- What are their main advantages and disadvantages?





Let's Explore!

A talk about Operating Systems



Instructions: Read the text below.

- Instructor:** Today, as part of the course we've been studying, we are going to discuss the main characteristics of multi-tasking, multi-processor, and distributed operating systems. Jay, please explain to us multi-tasking OS.
- Jay:** Sure, Multitasking Operating Systems are an extension of multiprogramming systems. These OS use their CPU for many jobs, it gives the possibility of hosting many users. Using multitasking OS would give users the perception of having their own computer.
- Instructor:** What have you found out about multiprocessor systems?
- Jay:** I've read multiprocessor systems may share the same bus and sometimes peripheral devices. There are two main types: the symmetric and asymmetric OS. The symmetric OS would use an identical copy of the OS and share the memory. The asymmetric OS would assign each processor a specific task, scheduling master and slave processors, it is more used among large systems.
- Instructor:** What about distributed operating systems?
- Jay:** They distribute the computation among many physical processors. All processors have a separate memory and communicate through buses, which make them have high speed systems. One of the benefits of this type of OS is that it provides file sharing and runs independently from other computers on the network.
- Instructor:** That's correct. You've talked appropriately about the OS studied in the class. They're all designed with different purposes and work with different ways. Next class we'll continue with some other types of OS. It would give us a wider perspective in the software industry.



Let's Practice!

Jay explains what multi-tasking operating systems are.

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

given / multiprogramming / have / used / have / has

Multitasking OS _____ been _____ systems. These OS _____
_____ their CPU for many jobs and It _____ the possibility of
hosting many users.



Let's Practice!

Jay describes what multi-processor operating systems are.

Instructions: Proofread the sentences. Correct the mistakes.

1. Multiprocessor systems woulded give the impression that they share the same bus and peripheral devices.
2. The multiprocessor systems would lets us think that there are two main types.
3. The symmetric OS would gives the idea that it uses an identical copy of the OS.
4. The symmetric OS would appears to share the memory of the OS system.
5. The asymmetric OS would gave the feeling that it assigns each processor a specific task.



Extension Activity

Now it's your turn to compare.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find information about multi-tasking operating systems, multi-processor operating systems and distributed operating systems.
- Focus on the benefits each one the systems provide.
- Organize the most important information in a Venn diagram.
- Give an oral presentation to explain to your partners the benefits of each one of the systems.

Self-Assessment

Yes No

- I find further information about operating systems. ☐ ☐
- I understand specific details of the information found. ☐ ☐
- I organize the information in a Venn diagram. ☐ ☐
- I explain to my partners the benefits each system offers. ☐ ☐



Target Words

- **master processor** (noun): a drive configured to give the commands and act as the main control.
e.g. The asymmetric OS would assign each processor a specific task, scheduling master and slave processors.
- **peripheral** (adjective): related to the devices such as keyboard, mouse or printers connected to a computer.
e.g. I've read multiprocessor systems may share the same bus and sometimes peripheral devices.
- **slave processor** (noun): a drive configured to be controlled by another device.
e.g. The asymmetric OS would assign each processor a specific task, scheduling master and slave processors.
- **file sharing** (noun): distribution of files among different users.
e.g. One of the benefits of this type of OS is that it provides file sharing and runs independently from other computers on the network.
- **host** (verb): to receive or serve as a host.
e.g. These OS use their CPU for many jobs, it gives the possibility of hosting many users.
- **job** (noun): in OS, it is a unit of work given to the Operating System.
e.g. These OS use their CPU for many jobs, it gives the possibility of hosting many users.



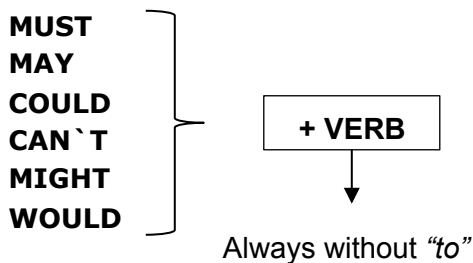
Learning Points

Modal verbs for speculating- Would

You can use the **modal verbs of speculation** to make guesses about the future or the present. They help you express the degrees of certainty in a sentence.

The **modal verbs of speculation** show the probability of a situation from the speaker's perspective. In this order of ideas, the speaker might be either wrong or right about what he/she is speculating.

The **modal verbs used to speculate** depend on how certain the speaker is regarding the sentence uttered. Take a look at the following modal verbs.



You need to remember modal verbs never take "s" in the third person of the singular *He/She/It*.

In the case of "would", it expresses what you think will happen:

Example:

Using a multi- tasking operating system would give the impression that everyone is sharing one computer at the same time.

Examples

- *Using multitasking OS would give users the perception of having their own computer.*
- *The symmetric OS would use an identical copy of the OS and share the memory.*

Multi-tasking, Multi-processor and Distributed Operating systems

Multi-tasking, Multi-processor and Distributed are types of Operating Systems with some differences and similarities.

Multi-tasking Operating Systems are an extension of multiprogramming systems. It allows many users from different locations to use a computer at the same time. It gives users the illusion of having their own machine, and allows duplication of software.





Learning Points

Multi-processor systems have more than one process in close communication, they share the same computer bus, memory and the clock. These types of systems may also share peripheral devices they've increased reliability.

Examples

- *I've read multiprocessor systems may share the same bus and sometimes peripheral devices.*
- *They distribute the computation among many physical processors. All processors have a separate memory and communicate through buses.*